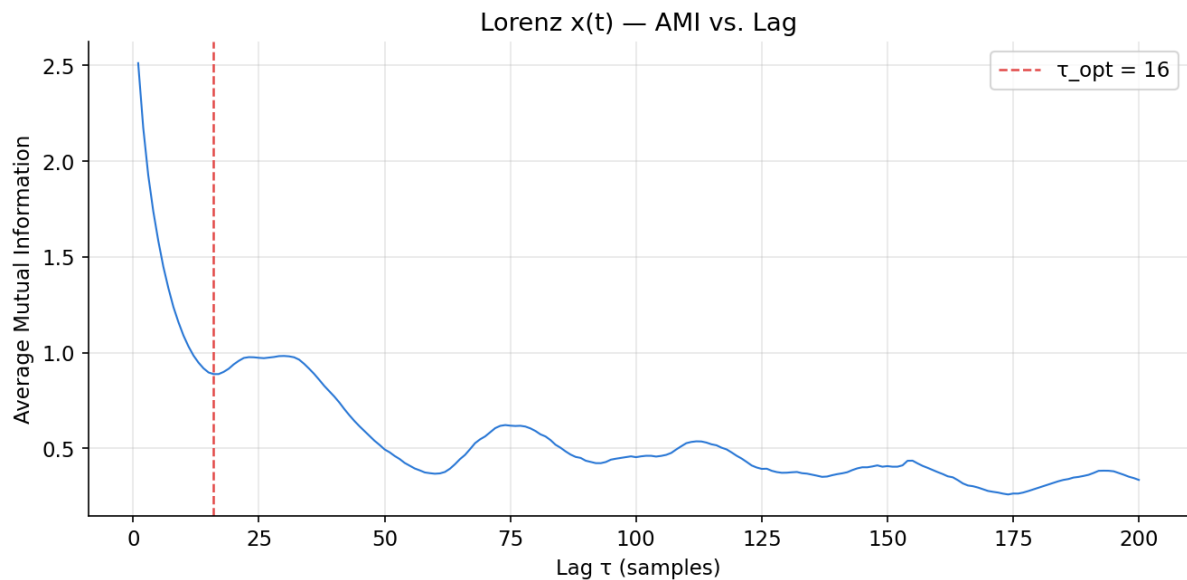


Name: Rana Mohamed Ahmed Shaqr

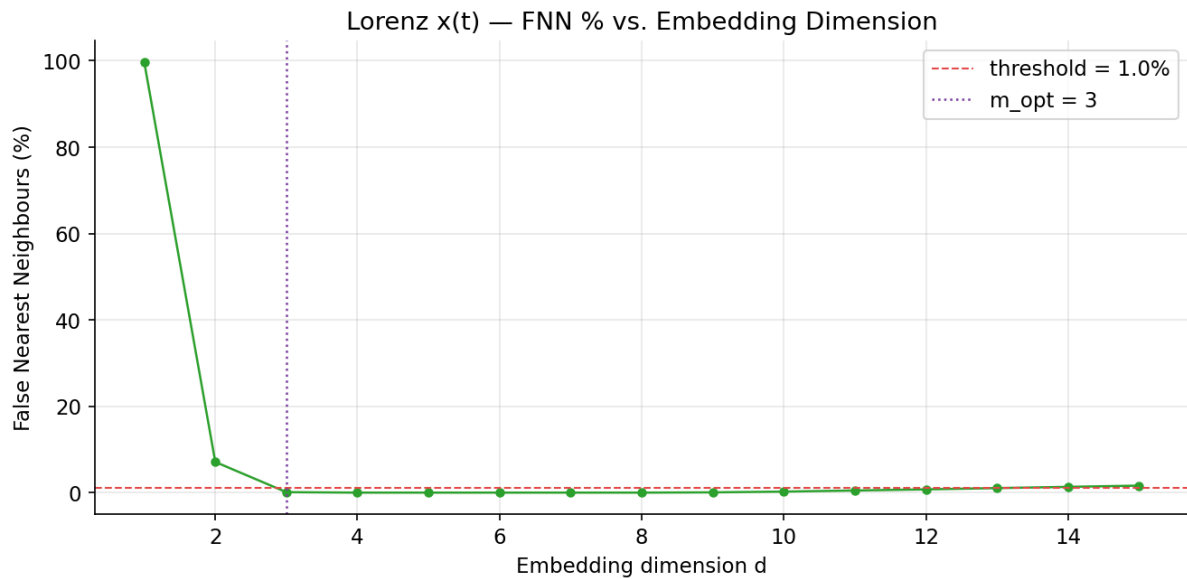
Group: LLM

Task 2

Part A:



When lag is 1-5 the AMI is largest as data is redundant. As τ increases, AMI decreases, meaning the shifted copy becomes less predictable from the original. First local minimum right at $\tau=16$ (at the edge of the expected 10–17 range). Enough time has passed that the two copies are no longer redundant, but not so much time that they've become statistically unrelated.



FNN starts very high at $d=1$ (the series folded onto one dimension has lots of fake crossings), drops sharply at $d=2$, and is already at 0% by $d=3$ so the smallest dimension clearing the 1% threshold is $m=3$, matching the lower bound of Lorenz's minimum embedding (3, since the true system genuinely lives in 3-D phase space). Takens' reconstruction, using only the single scalar $x(t)$ series, successfully recovered an attractor requiring no more coordinates than the real underlying system needs.

Part B:

B.2: Full spectrum from the known ODE (Wolf's GSR method)

Using the same Lorenz parameters and initial condition ($\sigma=10$, $\rho=28$, $\beta=8/3$, $IC=(1,1,1)$, $dt=0.01$, RK45), with `transient_steps=1000`, `n_steps=30000`, `ortho_steps=20`:

```
λ = (0.9003, 0.0042, -14.5633) nats/time
    = (1.2989, 0.0060, -21.0104) bits/time
```

As a validity check, the sum of the spectrum should equal $-(\sigma+1+\beta) = -13.6667$ for any Lorenz parameter set. The computed sum is -13.6588 , agreeing to within 0.06%.

The second exponent (0.0042 nats/time) is the theoretical zero exponent that every continuous-time autonomous flow has along the direction of motion itself. It is treated as marginal rather than positive. With exactly one exponent clearly positive, one marginal, and one strongly negative, the system is classified as **chaotic**, consistent with Lorenz's known 3-dimensional phase space. $\lambda_1 = 0.9003$ nats/time also matches the value for these parameters (≈ 0.906 nats/time) to within 0.6%.

B.3: λ_1 from a scalar time series (Wolf's tracking method)

Using the same $\tau=16$, $m=3$ embedding parameters, on a 15,000-sample $x(t)$ series:

$$\lambda_1 = 1.4584 \text{ nats/time} = 2.1041 \text{ bits/time}$$

300 segments were tracked over 149 physical time units.

Method	λ_1 (nats/time)	λ_1 (bits/time)
ODE / Wolf GSR (ground truth)	0.9003	1.2989
Time series / Wolf tracking	1.4584	2.1041
Relative difference	—	62.0%

Discussion of the discrepancy

Both methods agree that λ_1 is clearly positive, confirming chaotic behaviour by either approach, but the time-series estimate overshoots the ODE-based ground truth by roughly 62%. Segment-level diagnostics point to a likely cause: 292 of the 300 tracked segments were cut off by hitting `evolve_cap` (50 steps) rather than stopping naturally once their separation exceeded the divergence threshold meaning most segments were forcibly truncated before necessarily settling into the clean exponential-growth regime the $\log_2(d_{\text{final}}/d_0)$ formula assumes. This, combined with the inherent sensitivity of Wolf's tracking method to its distance and angle-weighting parameters (not tuned specifically for this attractor), is consistent with the usual causes for this kind of discrepancy: finite series length, embedding quality, and tracking-parameter sensitivity. The ODE-based GSR spectrum should be treated as the reliable reference value; the time-series method demonstrates that a reasonably accurate λ_1 can still be recovered from a single scalar signal alone, at the cost of a substantially noisier estimate.

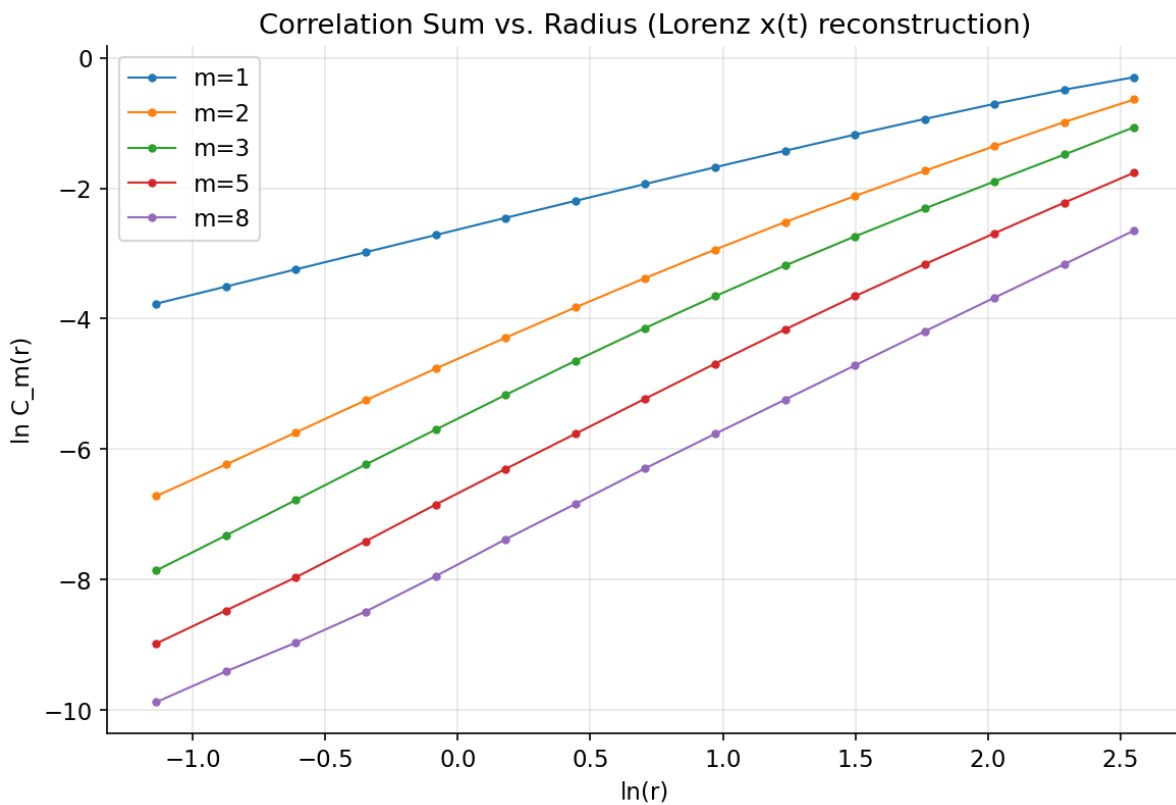
Part C:

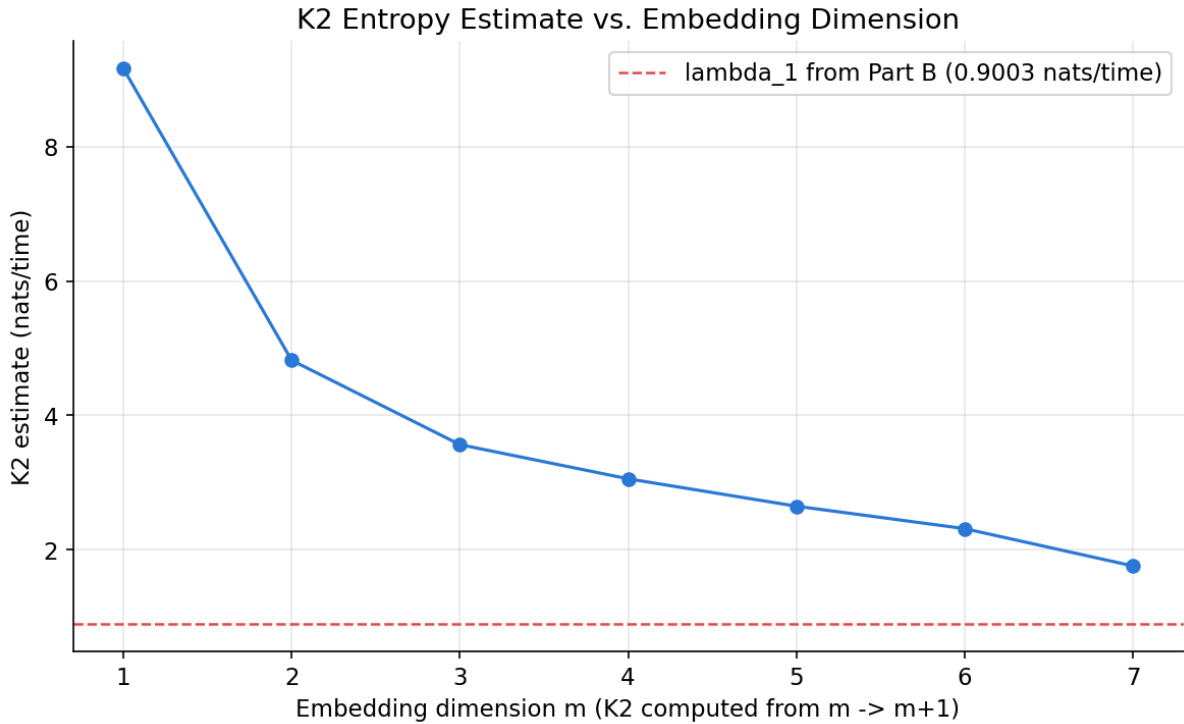
```

PS C:\Users\user\Downloads\Rana_NU\Task 2> python run_partc.py
Series length after transient discard: 15000
Delta_t used in K2 formula (tau*dt): 0.160 time units
m=1: C(r) range = [2.30e-02, 7.43e-01]
m=2: C(r) range = [1.21e-03, 5.28e-01]
m=3: C(r) range = [3.85e-04, 3.45e-01]
m=4: C(r) range = [2.05e-04, 2.38e-01]
m=5: C(r) range = [1.26e-04, 1.72e-01]
m=6: C(r) range = [8.63e-05, 1.29e-01]
m=7: C(r) range = [6.38e-05, 9.50e-02]
m=8: C(r) range = [5.12e-05, 7.09e-02]
Saved: C_logC_vs_logr.png
K2 estimate from m=1->m+1=2: 9.1613 nats/time (std over scaling region: 2.3173)
K2 estimate from m=2->m+1=3: 4.8212 nats/time (std over scaling region: 0.6581)
K2 estimate from m=3->m+1=4: 3.5659 nats/time (std over scaling region: 0.2854)
K2 estimate from m=4->m+1=5: 3.0571 nats/time (std over scaling region: 0.2181)
K2 estimate from m=5->m+1=6: 2.6496 nats/time (std over scaling region: 0.1033)
K2 estimate from m=6->m+1=7: 2.3157 nats/time (std over scaling region: 0.0393)
K2 estimate from m=7->m+1=8: 1.7608 nats/time (std over scaling region: 0.0358)
Saved: C_K2_vs_m.png

===== SUMMARY =====
K2 estimates by m: [9.1613 4.8212 3.5659 3.0571 2.6496 2.3157 1.7608]
K2 at largest m tested: 1.7608 nats/time
Pesen identity reference (lambda_1 from Part B): 0.9003 nats/time
PS C:\Users\user\Downloads\Rana_NU\Task 2>

```





Using the same scalar Lorenz series ($\tau=16$, m up to 8), correlation sums $C_m(r)$ were computed at embedding dimensions $m=1$ through $m=8$, with a Theiler window of $\tau \cdot m$ excluding temporally correlated pairs.

Correlation sum curves. The $\log C_m(r)$ vs. $\log(r)$ curves are approximately linear over the middle portion of the radius range for every tested dimension, confirming the power-law scaling assumption $C_m(r) \propto r^{D2} \cdot e^{(-m\Delta t K2)}$ underlying the K2 estimator is reasonable for this dataset. The curves shift downward and become closer to parallel as m increases, consistent with the attractor progressively unfolding (same as FNN)

$k2$ is estimated by using $\Delta t = \tau \cdot dt = 0.16$ time units, and averaging $-(1/\Delta t) \cdot \ln[C_m(r)/C_{m+1}(r)]$ over the middle (scaling) portion of the radius grid.

The estimates decrease, the *size of each successive drop* also shrinks ($7.34 \rightarrow 1.26 \rightarrow 0.51 \rightarrow 0.41 \rightarrow 0.33 \rightarrow 0.56$ nats/time), rather than staying constant or growing. this decelerating-decrease pattern is the expected signature of convergence toward a finite value, low-dimensional deterministic chaos, not noise. The standard deviation of the K2 estimate across the scaling region also drops ($2.317 \rightarrow 0.036$), indicating the estimate becomes markedly more stable and radius-independent and more trustworthy at higher m .

Pesin's identity predicts K should approach ≈ 0.90 nats/time as $m \rightarrow \infty$. At $m=8$ the estimate (1.761 nats/time) is still roughly double this reference value, but the clear decelerating downward trend is consistent with genuine convergence toward it, limited here by the finite series length (15,000 samples) and the modest maximum embedding

dimension tested ($m=8$). A longer series and/or higher m would be expected to bring the estimate closer to the 0.90 reference.

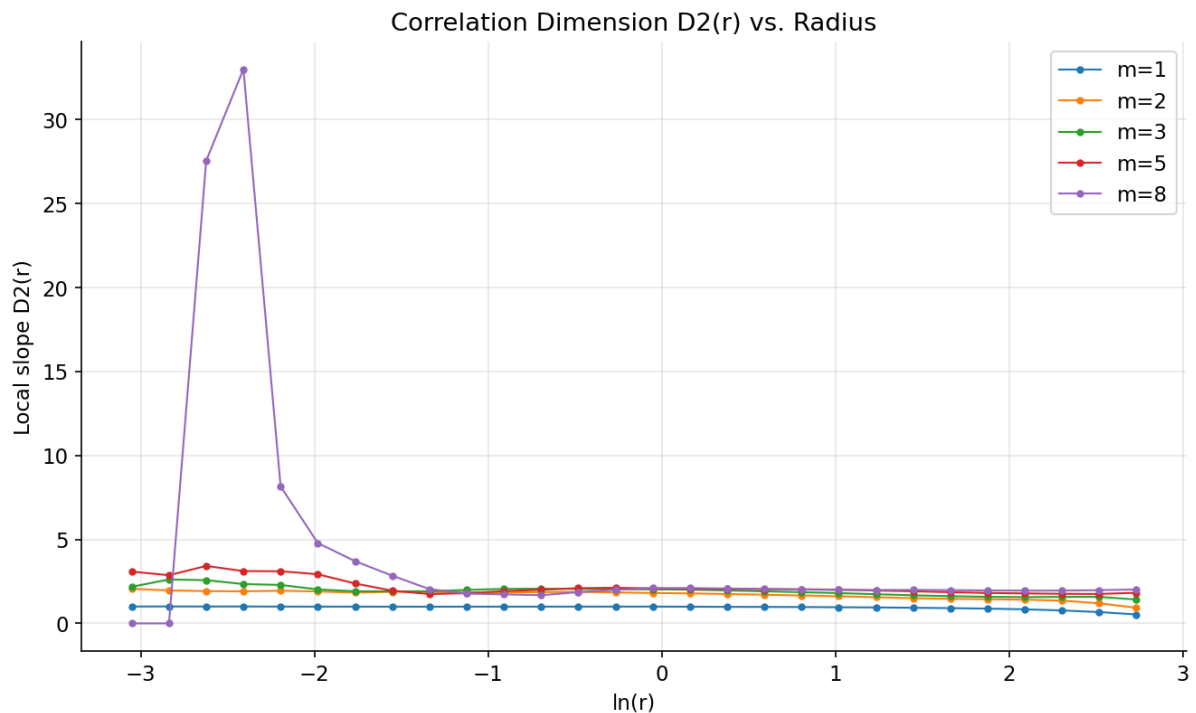
Part D:

```

PS C:\Users\user\Downloads\Rana_NU\Task 2> python run_partd.py
Series length after transient discard: 15000
Estimated attractor diameter (m=3 embedding): 38.135
m=1: C(r) computed, range [2.74e-03, 9.14e-01]
m=2: C(r) computed, range [1.98e-05, 7.57e-01]
m=3: C(r) computed, range [3.74e-06, 6.03e-01]
m=4: C(r) computed, range [1.25e-06, 4.62e-01]
m=5: C(r) computed, range [3.57e-07, 3.56e-01]
m=6: C(r) computed, range [1.00e-12, 2.71e-01]
m=7: C(r) computed, range [1.00e-12, 2.08e-01]
m=8: C(r) computed, range [1.00e-12, 1.58e-01]
Saved: D_D2_vs_r.png
m=1: D2 = 0.990 (std over scaling region: 0.012)
m=2: D2 = 1.783 (std over scaling region: 0.084)
m=3: D2 = 1.979 (std over scaling region: 0.083)
m=4: D2 = 2.018 (std over scaling region: 0.065)
m=5: D2 = 2.020 (std over scaling region: 0.087)
m=6: D2 = 2.000 (std over scaling region: 0.135)
m=7: D2 = 1.965 (std over scaling region: 0.150)
m=8: D2 = 1.945 (std over scaling region: 0.149)
Saved: D_D2_vs_m.png

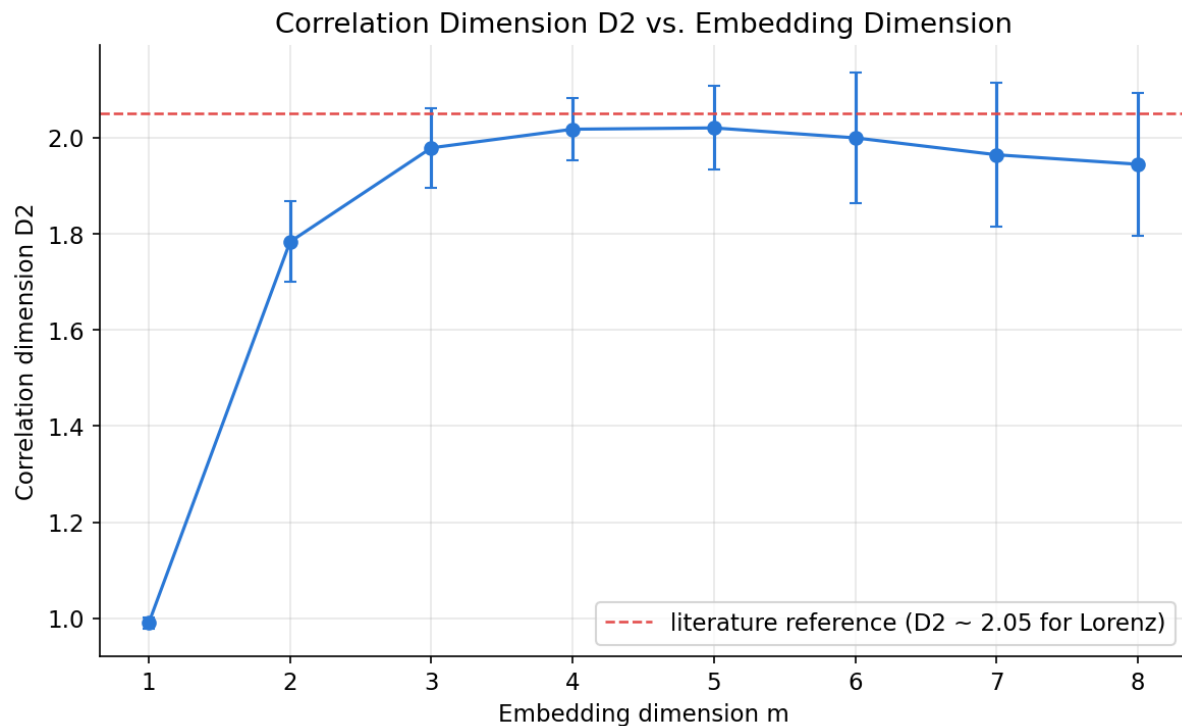
===== SUMMARY =====
D2 estimates by m: [0.99 1.783 1.979 2.018 2.02 2. 1.965 1.945]
D2 at largest m tested (m=8): 1.945
Literature reference for Lorenz (sigma=10, rho=28, beta=8/3): D2 ~ 2.05
PS C:\Users\user\Downloads\Rana_NU\Task 2>

```



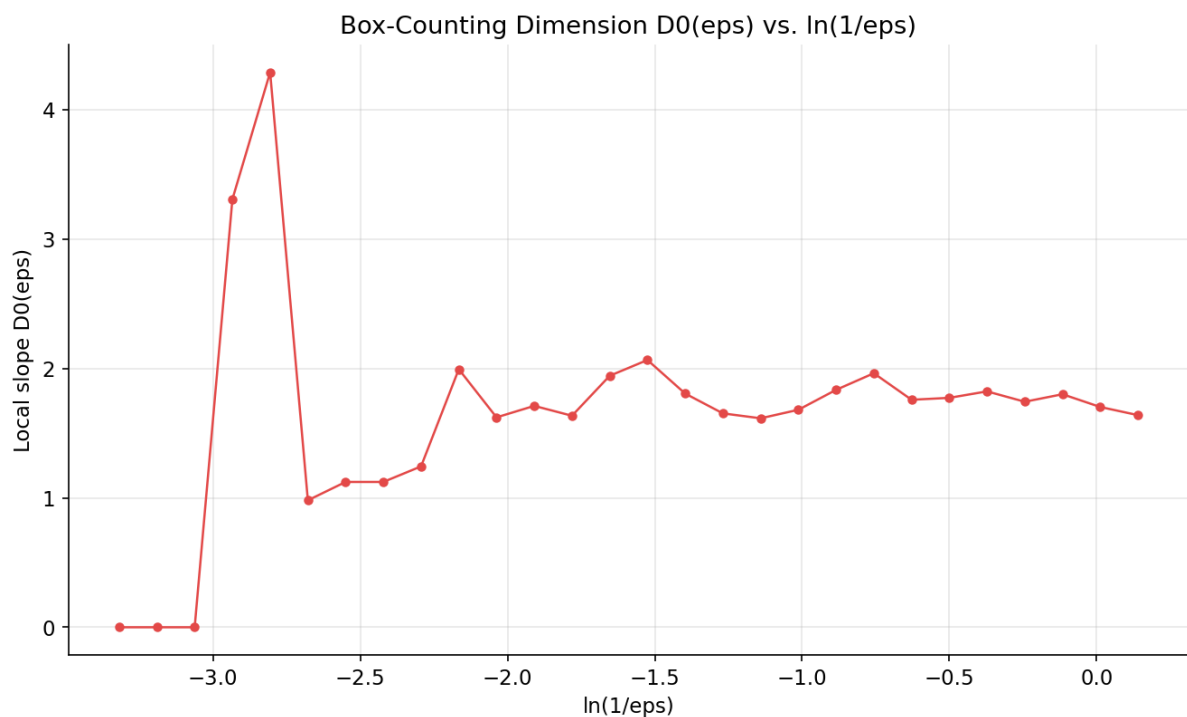
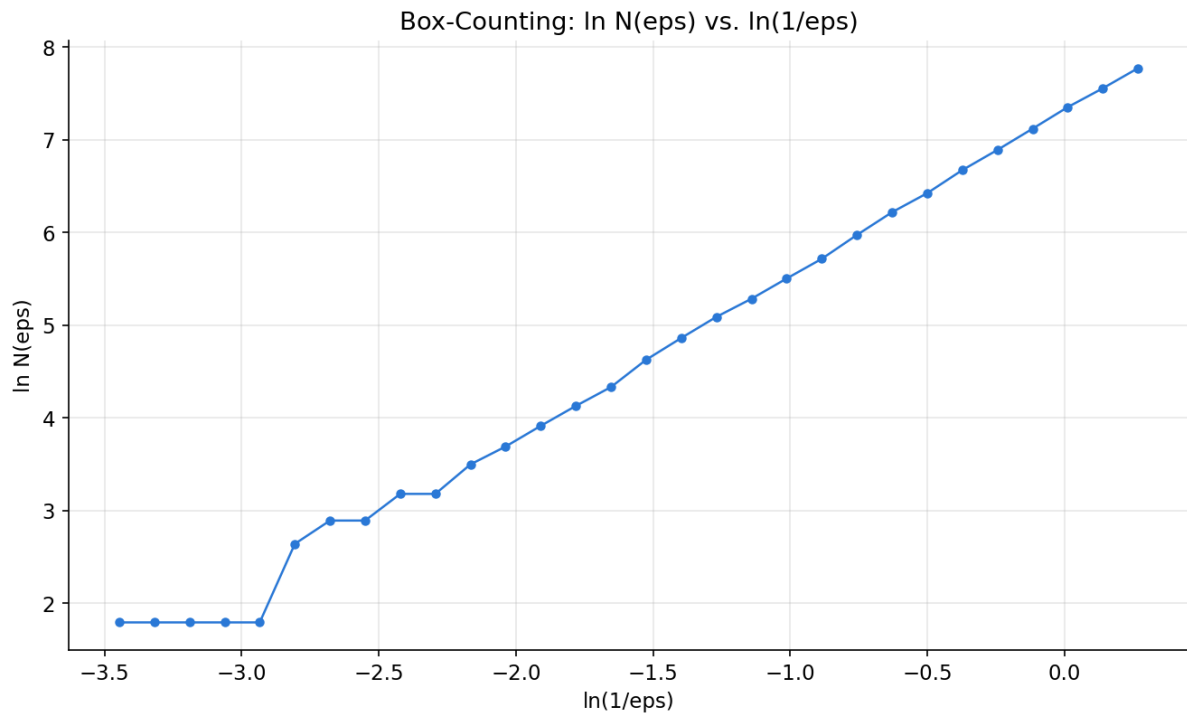
At low m , the $D_2(r)$ curves are essentially flat, hovering near $D_2 \approx 1-2$ across the whole

radius range. At $m=8$, there's a sharp artificial spike (up to ~ 33) at very small r , this is a numerical artifact from too few point-pairs being found at tiny radii and high dimension, not real structure. All curves converge onto a shared flat plateau around $D_2 \approx 2$ in the middle-to-large radius range.



D_2 vs. embedding dimension: D_2 rises sharply from ~ 0.99 at $m=1$ to ~ 1.98 - 2.02 by $m=3$ - 5 , then plateaus, then drifts slightly down (to 1.945) by $m=8$ with widening error bars (std). The plateau around $m=3$ - 5 ($D_2 \approx 2.0$ - 2.02) is the reliable saturation value, closely matching the reference of 2.05 confirming the attractor is a low-dimensional fractal object; the decline at higher m reflects data sparsity, not a real change in the attractor's dimension.

Part E:



Theory predicts $D_0 \geq D_2$ for the same attractor; The result ($D_0 = 1.762$) is below $D_2 \approx 2.02$, the opposite of the expected ordering. This is a limitation of box-counting rather than evidence of an error: box-counting only records whether each grid cell is occupied, discarding the finer pairwise-distance information that the correlation sum uses, so it requires more data to converge to its true value. At a finite sample size (15,000 points here), box-counting systematically underestimate D_0 relative to density-weighted measures like D_2 ; consistent with both the direction and rough magnitude of the discrepancy found here.

